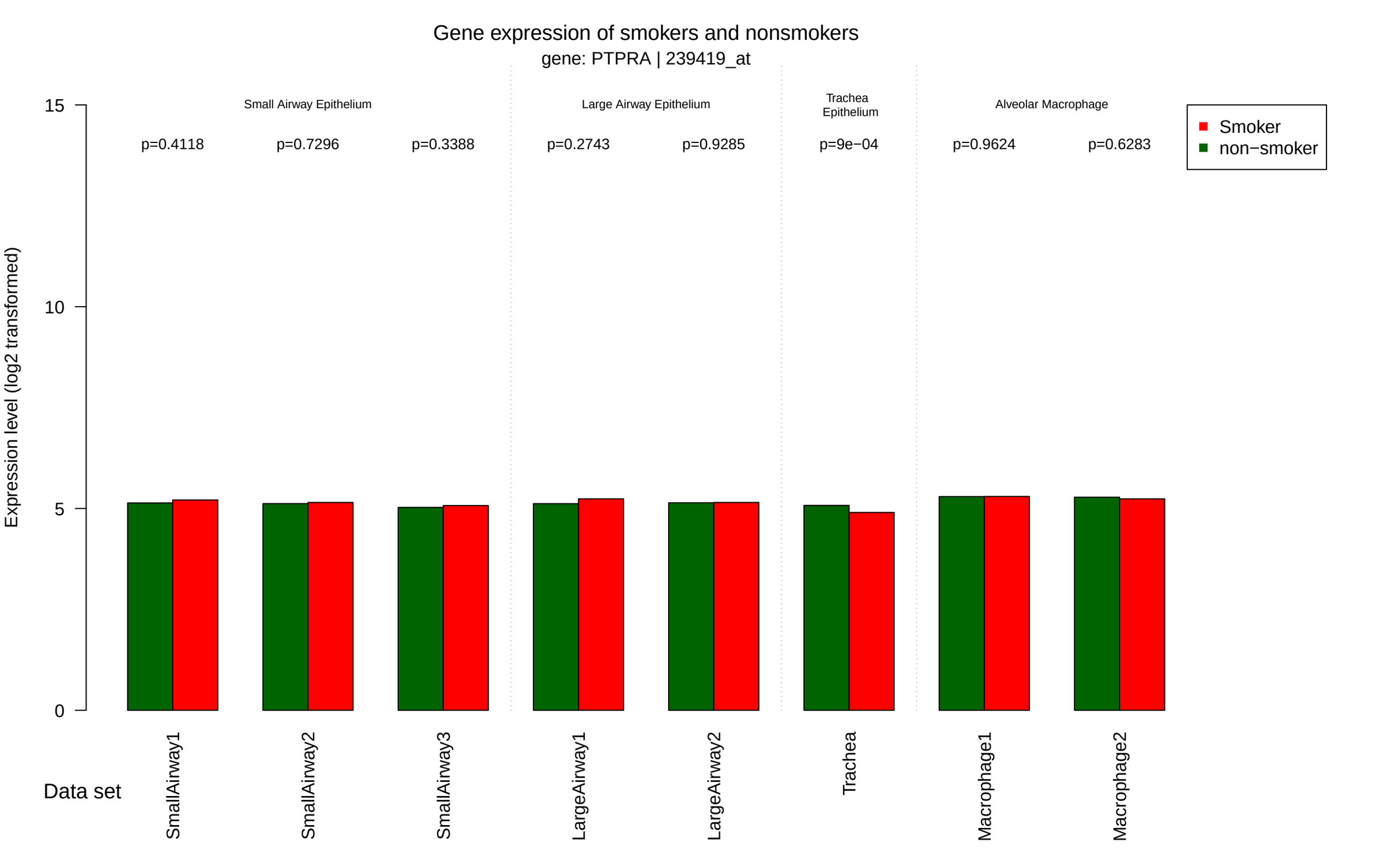
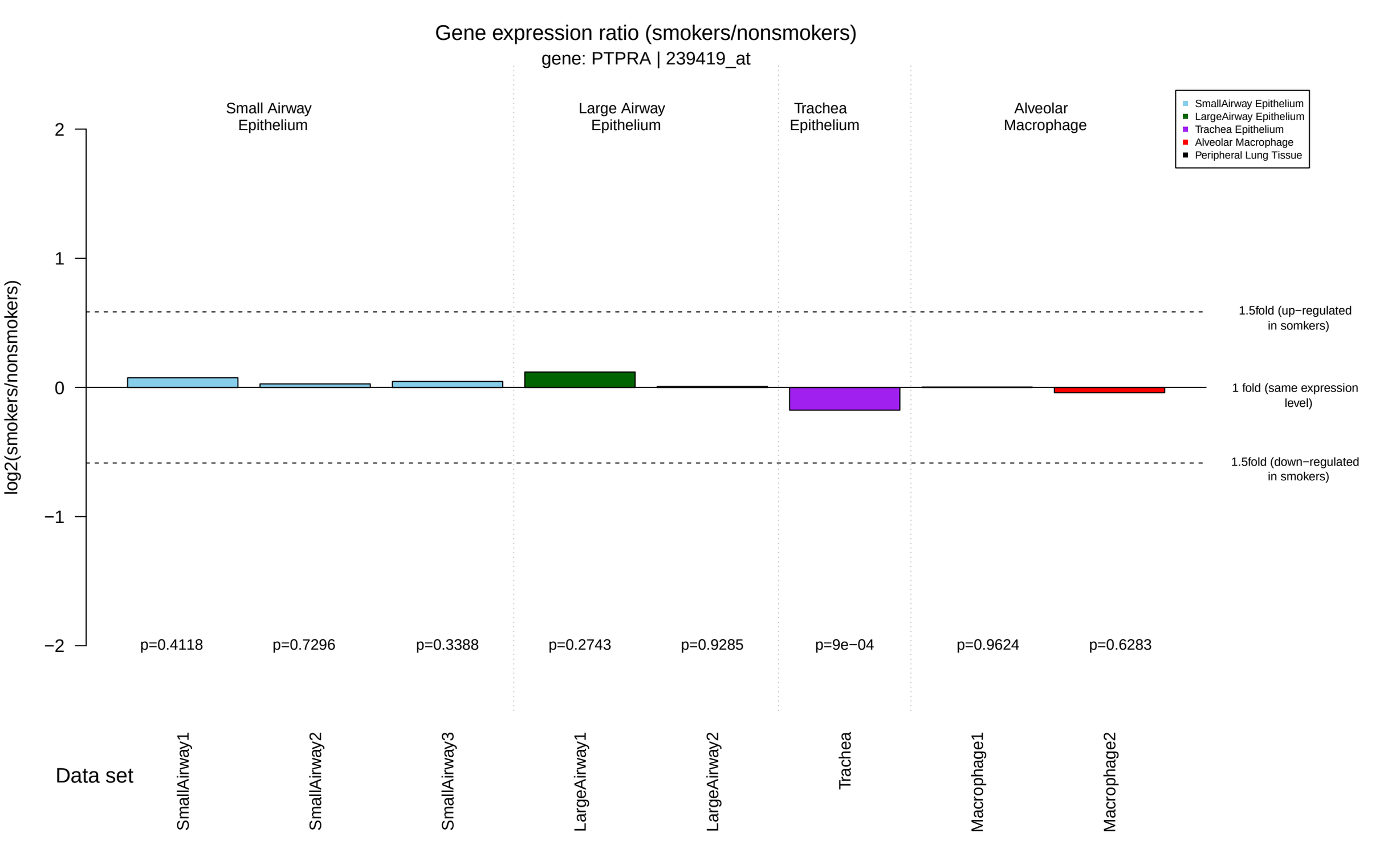
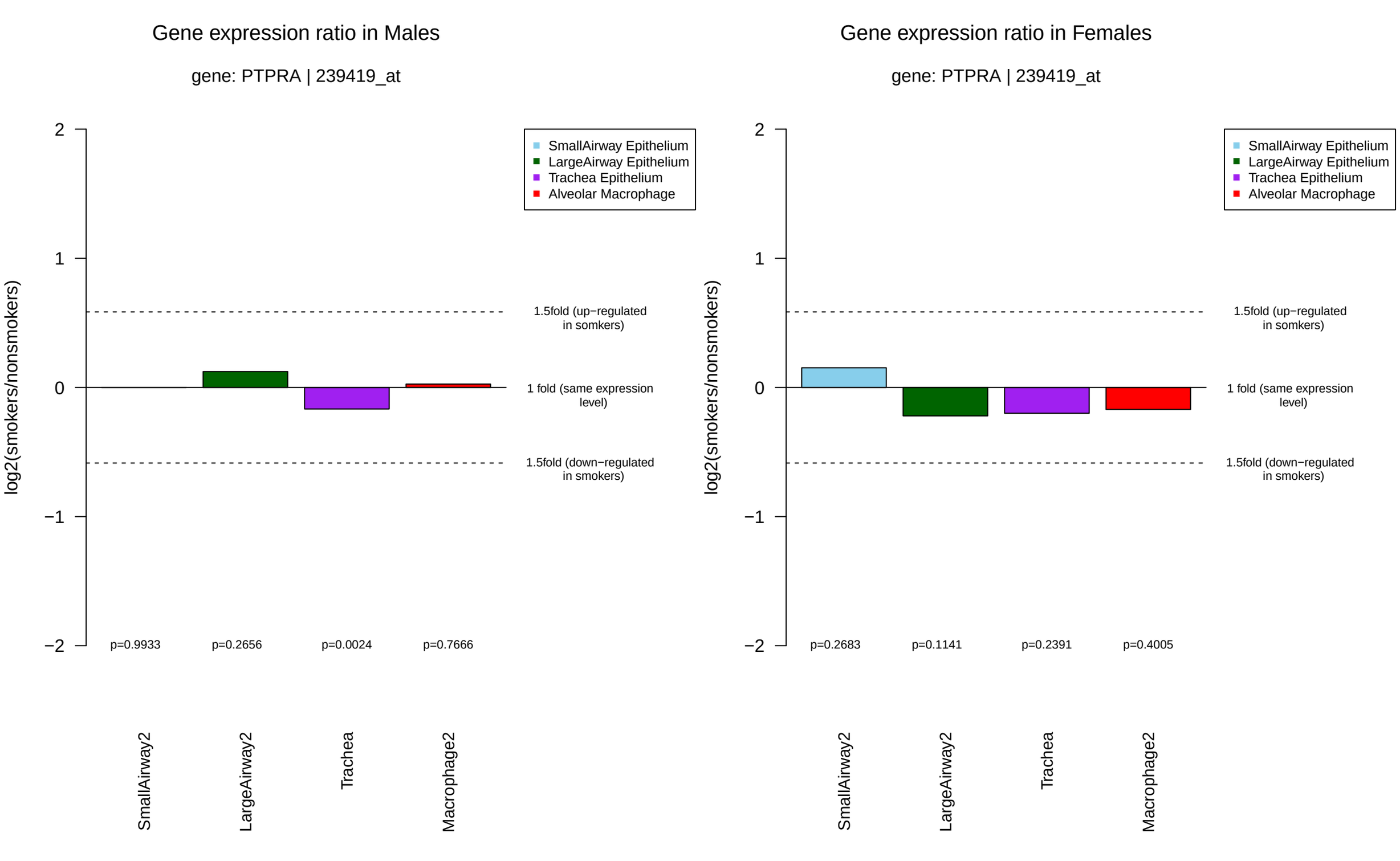
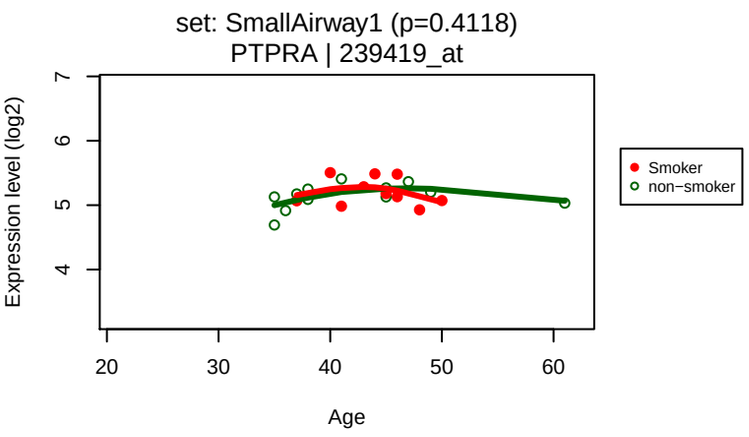
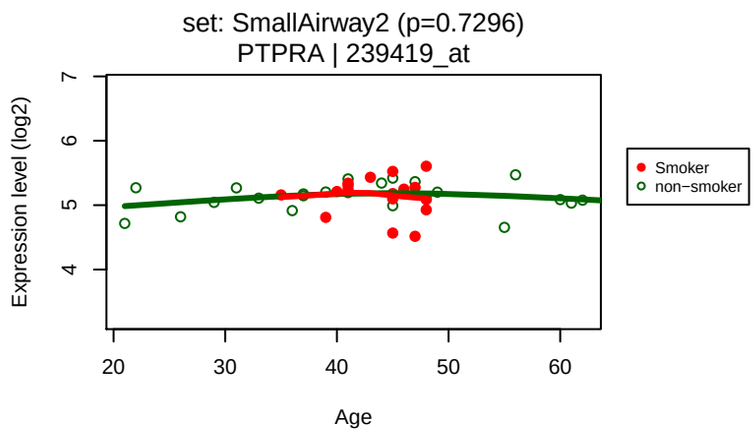
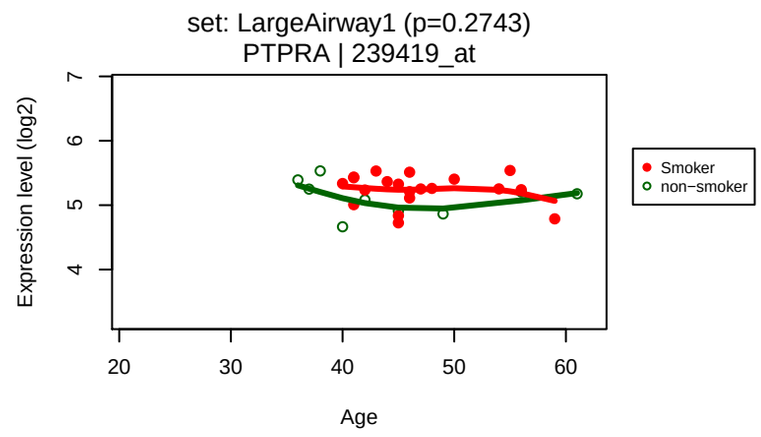

Gene expression of smokers and nonsmokers
gene: PTPRA | 239419_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4118
p=0.7296
p=0.3388
p=0.2743
p=0.9285
p=9e−04
p=0.9624
p=0.6283
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

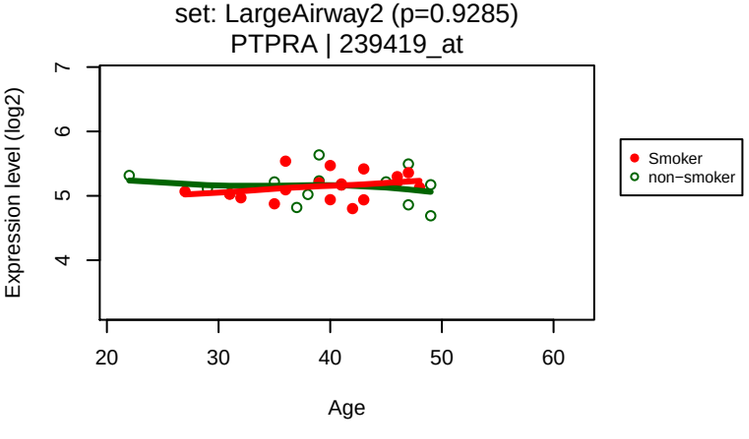

Gene expression ratio (smokers/nonsmokers)
gene: PTPRA | 239419_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4118
p=0.7296
p=0.3388
p=0.2743
p=0.9285
p=9e−04
p=0.9624
p=0.6283
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

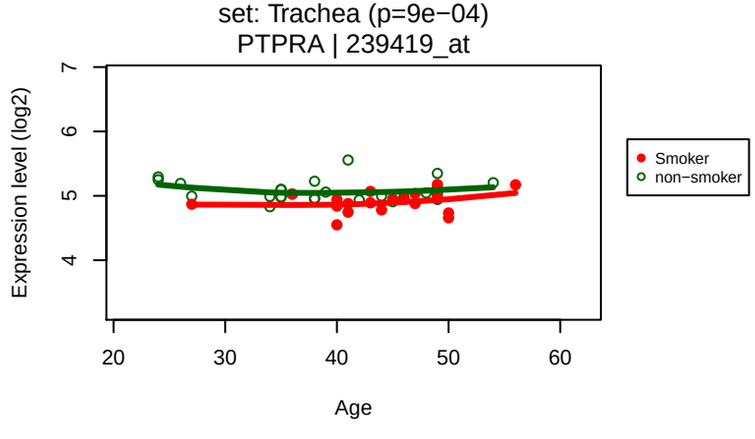

Gene expression ratio in Males
gene: PTPRA | 239419_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9933
p=0.2656
p=0.0024
p=0.7666
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPRA | 239419_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2683
p=0.1141
p=0.2391
p=0.4005
SmallAirway2
LargeAirway2
Trachea
Macrophage2

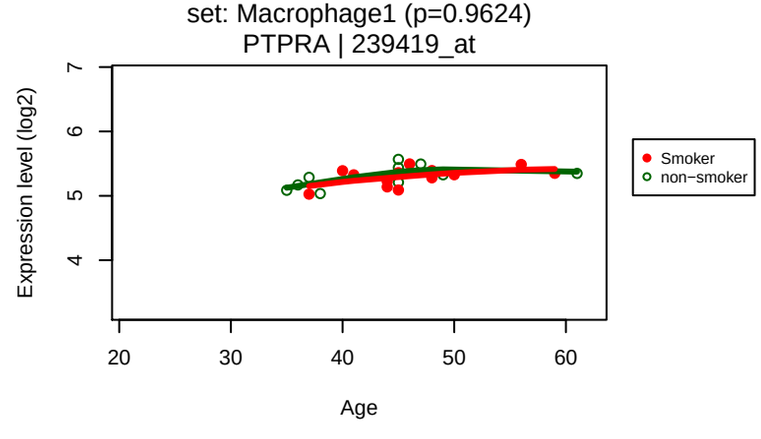

set: SmallAirway1 (p=0.4118)
PTPRA | 239419_at
Expression level (log2)
Age
Smoker
non−smoker

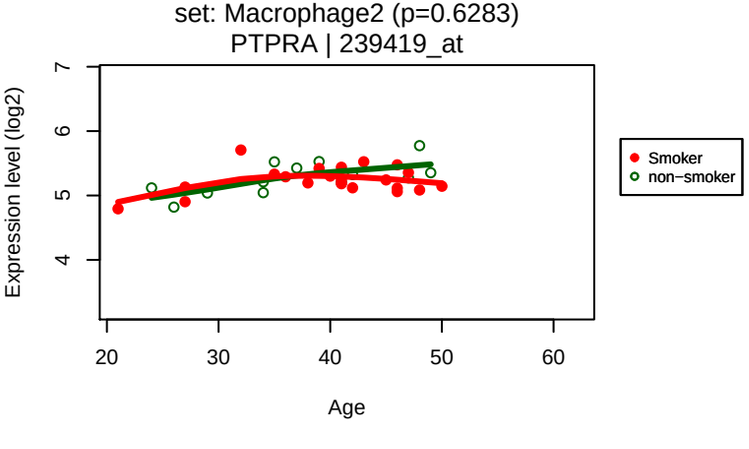

set: SmallAirway2 (p=0.7296)
PTPRA | 239419_at
Expression level (log2)
Age
Smoker
non−smoker

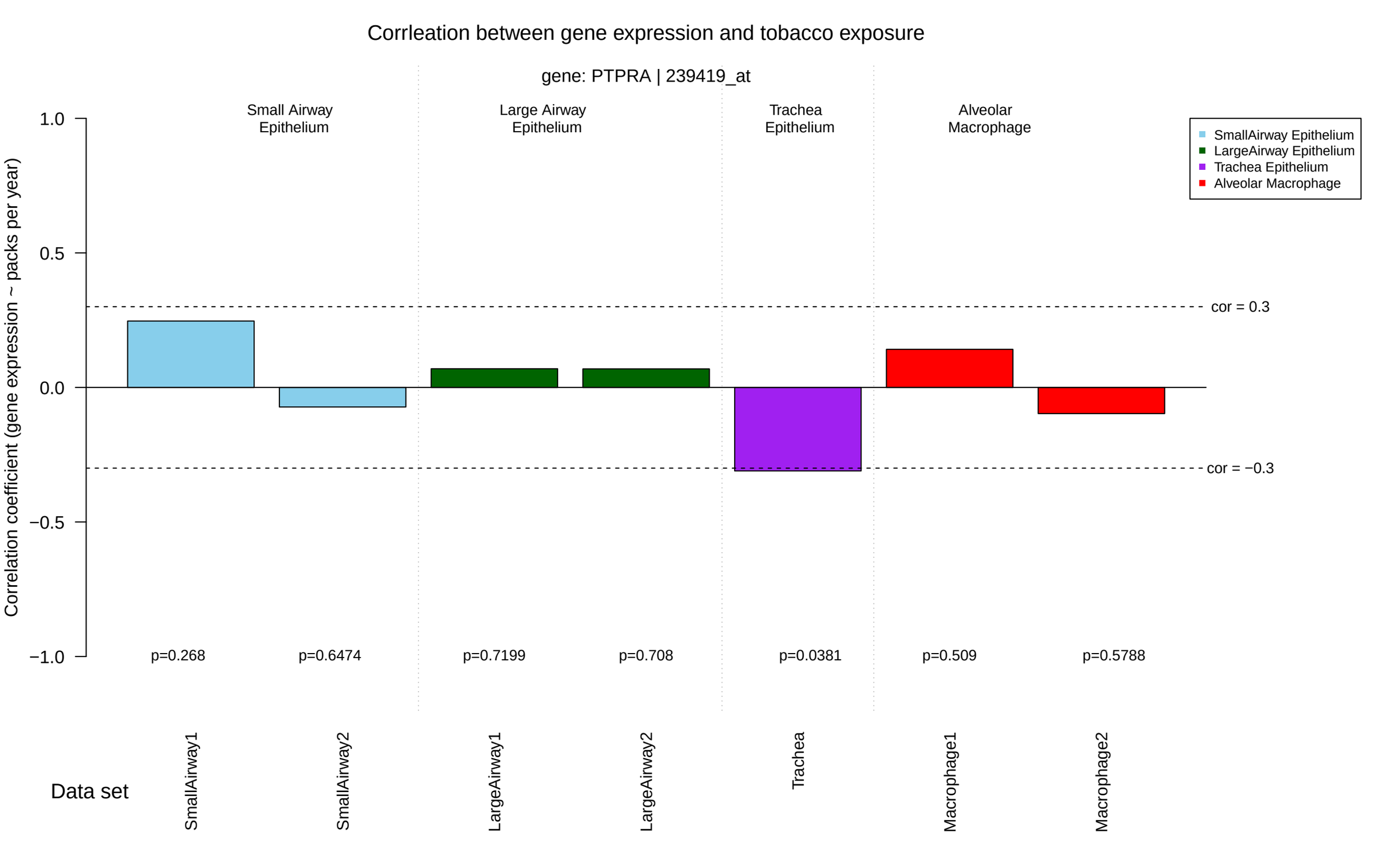

set: LargeAirway1 (p=0.2743)
PTPRA | 239419_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9285)
PTPRA | 239419_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=9e−04)
PTPRA | 239419_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9624)
PTPRA | 239419_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6283)
PTPRA | 239419_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPRA | 239419_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.268
p=0.6474
p=0.7199
p=0.708
p=0.0381
p=0.509
p=0.5788
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2